

Problem Set #1- Due Date 9/15/26

This problem set introduces you to tests of the Q-theory of investment and estimate investment-cash flow sensitivities using simple OLS regressions on 3 corporations found in Compustat. It is based on empirical work by Fazzari, Hubbard, and Petersen (1988, *Brookings Papers on Economic Activity*, hereafter FHP) and Erickson and Whited (2012, *Review of Financial Studies*). You will conduct a much more comprehensive assignment on this topic in Oliver Levine's portion of the class.

There are two parts to this assignment. First you will construct a panel of variables necessary to run FHP regressions of the form

$$\frac{I_{i,t}}{K_{i,t-1}} = \beta_0 + \beta_1 Q_{i,t-1} + \beta_2 CF_{i,t} + \epsilon_{i,t}$$

where $\frac{I_{i,t}}{K_{i,t-1}}$ is the ratio of capital expenditures to lagged property, plant, and equipment (PP&E), and $Q_{i,t-1}$ is the market to book value of assets, and $CF_{i,t}$ is the ratio of cash flow to lagged book assets for firm i at time t .¹

For a sample period of 1980-2008, we chose 3 corporations after creating a balanced panel, ranking all firms in Compustat by their size (PP&E), forming terciles, creating a subset of each tercile of firms which remain in that tercile over the length of the sample, and choosing one firm from each tercile subset. Specifically, the small sized firm we choose to focus on is EMS Technologies Inc (gvkey 4279), the medium sized firm is Atwood Oceanics Inc. (gvkey 1860), and the big sized firm is Raytheon Technologies Corp. (gvkey 10983). The variables in the excel spreadsheet are `gvkey` (firm identifier in Compustat), `tic` (firm ticker symbol), `ppeg` (property plant and equipment), `conml` (firm name), `inv_rate` (capital expenditures/lagged ppeg), `tobin_q_lag` (lagged Tobin's Q), `cash_flow` (cash flow), and `bin` (size tercile).

After creating the dataset from Comustat, you are to replicate all the remaining figures and tables. Figure 1 - 3 plots the investment rate, Q, and cash flow for the three firms. Figure 4 shows how the capital stocks of each firm evolves over time. Summary statistics for each firm are given in table 1.

¹This problem set uses the variable definitions given for the *second* set of regressions in the appendix of Erickson and Whited (2012) on page 1325. Note that some of these are not canonical in that there are alternative ways to reasonably measure some of these objects. The appendix to this problem set provides the set of filters used to construct the necessary data set. See the appendix for more detailed information.

Figure 1: EMS Technologies Inc.

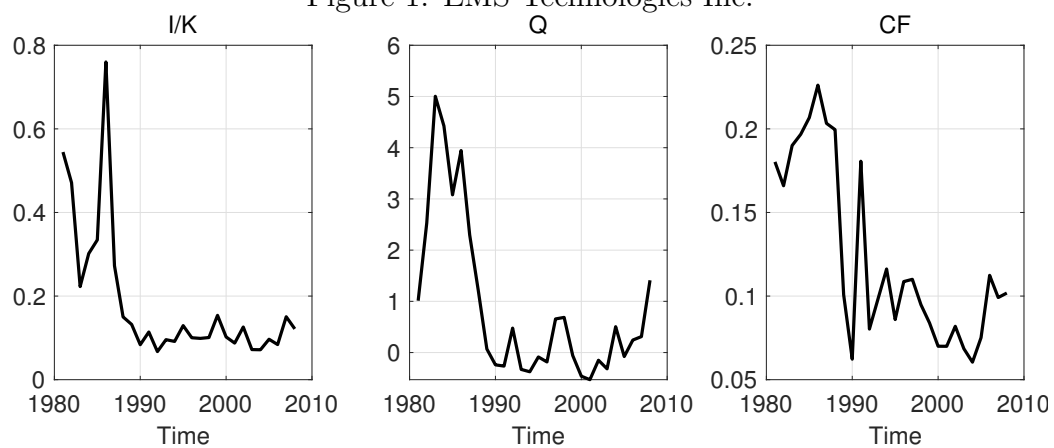


Figure 2: Atwood Oceanics Inc.

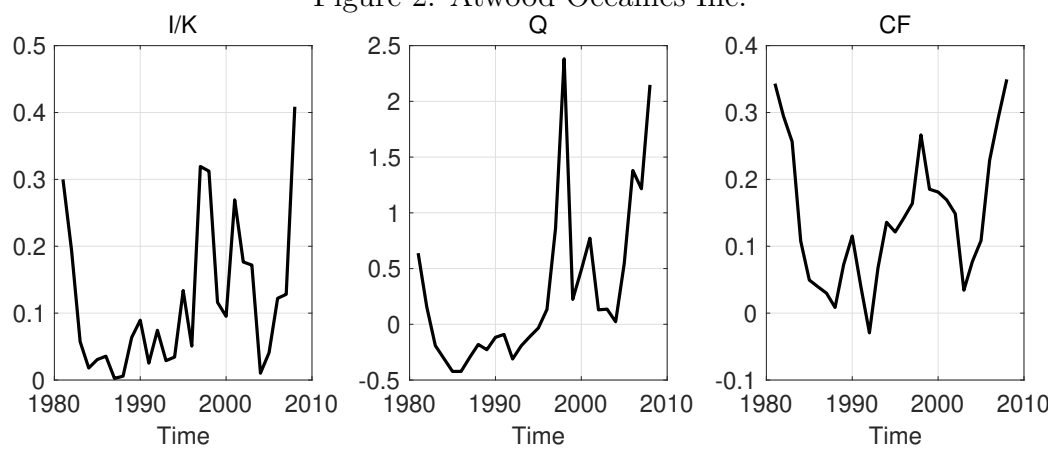


Figure 3: Raytheon Technologies Corp.

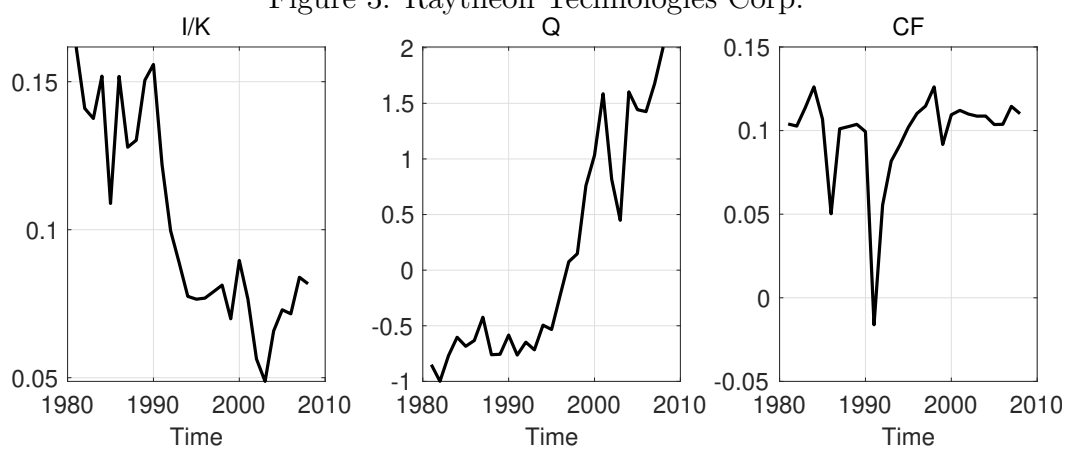


Figure 4: Capital Stock
 k

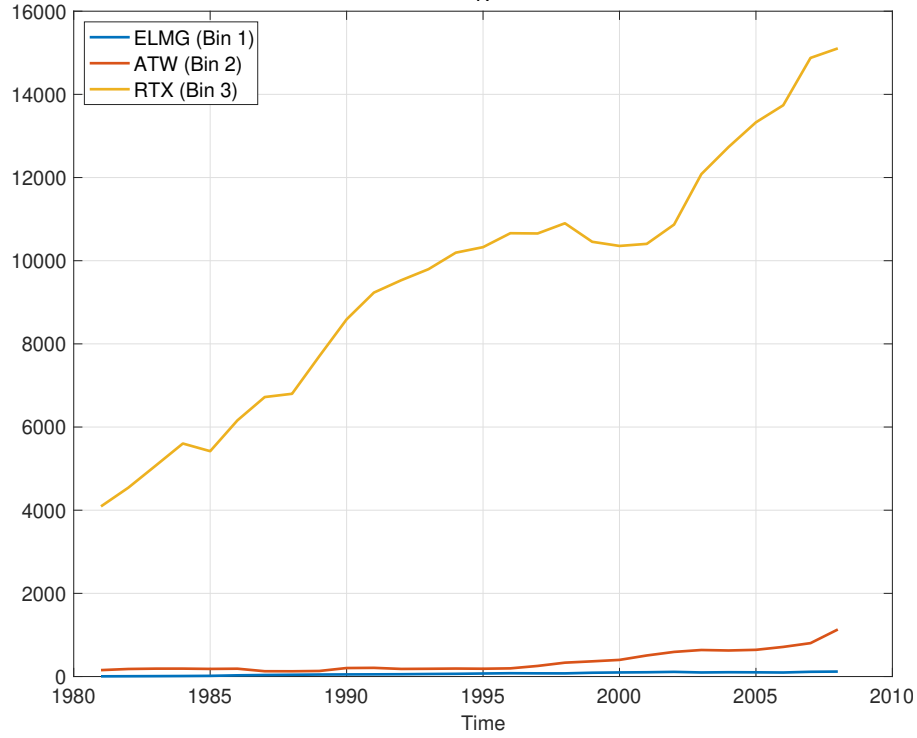


Table 1: Summary Statistics

	Mean	St. Dev
EMS Technologies Inc		
k	63.44	(37.05)
i/k	0.18	(0.16)
CF	0.12	(0.05)
Q	0.89	(1.57)
Atwood Oceanics Inc.		
k	343.63	(255.45)
i/k	0.12	(0.11)
CF	0.14	(0.10)
Q	0.30	(0.73)
Raytheon Technologies Corp.		
k	9296.61	(3211.94)
i/k	0.10	(0.03)
CF	0.10	(0.03)
Q	0.09	(0.97)

Regression results when β_2 is constrained to be zero (the prediction of Modigliani-Miller) and when β_2 is estimated are in tables 2 and 3, respectively.

Table 2: Regression Results (No Cash Flow)

	EMS Technologies Inc	Atwood Oceanics Inc.	Raytheon Technologies Corp.
	<i>i/k</i>	<i>i/k</i>	<i>i/k</i>
Q_{t-1}	0.07*** (0.03)	0.12*** (0.02)	0.06*** (0.02)
Observations	28	28	28
R^2	0.442	0.596	0.292

Table 3: Regression Results

	EMS Technologies Inc	Atwood Oceanics Inc.	Raytheon Technologies Corp.
	<i>i/k</i>	<i>i/k</i>	<i>i/k</i>
Q_{t-1}	0.03 (0.03)	0.09*** (0.02)	-0.02*** (0.00)
CF_t	1.64* (0.87)	0.31 (0.18)	0.03 (0.13)
Observations	28	28	28
R^2	0.552	0.637	0.457

Note that the order of β_2 is consistent with the predictions of FHP; specifically the smaller, more likely to be constrained firms, have higher cash flow sensitivity. Note further that cash flow (CF_t) is correlated with lagged Tobin's Q (Q_{t-1}), since Q is the present discounted value of the firm. By leaving cash flow in the error term (in Table 2) we violate one of the assumptions of OLS that the error term is uncorrelated with our regressors and hence the estimate on Q is inconsistent and biased which shows up in the fact that β_1 falls from Table 2 to Table 3.

1 Appendix

The data uses the following filters (each observation must satisfy all of the following criteria):

1. `(indfmt == "INDL" & datafmt == "STD" & popsrc == "D" & consol == "C")`

These conditions ensure that gvkey-fyear uniquely identify each observation.

2. `(at>0)`

Positive book assets.

3. `(sic < 6000 | sic > 6999)`

No financial firms.

4. `(sic < 4900 | sic > 4999)`

No utilities.

5. `(gvkey != "" & fyear != .)`

No missing identifiers.

6. `(fic == "USA")`

Only US firms. Note that some papers use US and Canada; others use only US.

7. `(prcc_f != . & csho != .)`

Drop missing stock market data. Note that the CRSP data is more accurate than Compustat for stock price data. It's a good idea to use CRSP when possible, but many in the corporate finance literature use the Compustat to construct market cap. You can use the CRSP-Compustat merged database to combine the two: the matching has been done.

1.1 Variable Definition

- `capx`: Capital Expenditures
- `ppeg`: Property Plant and Equipment, Gross Total
- $I_t/K_{t-1} = \text{capx}_t/\text{ppeg}_{t-1}$
- `dltt`: Long Term Debt
- `dlc`: Short Term Debt

- `prcc`: Stock Price
- `csho`: Shares Outstanding
- `at`: Total Assets
- $Q_t = (\text{dltt}_t + \text{dlc}_t + \text{prcc}_t \times \text{csho}_t - \text{at}_t) / \text{ppeg}_t$
- `ibc`: Income Before Extraordinary Items
- `dpc`: Depreciation and Amortization
- $CF_t = (\text{ibc}_t + \text{dpc}_t) / \text{at}_{t-1}$

Data is Winsorized at 99/1 percentiles. Variable definitions come from Erickson and Whited (2012).

1.2 Constructing Bins

- Generate a balanced panel.
- For each year, sort firms into terciles based on their size (`ppeg`).
- Restrict sample to firms that stay in the same tercile over the sample.